

John Finnegan

Staff Engineer

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PROFILE

Software Engineer with experience building **scalable fintech and developer platforms** across wealth management and software quality domains, including work at Addepar and SmartBear. Strong background in **cloud-native architecture, CI/CD automation, portfolio data pipelines, and API-driven microservices**, delivering high-performance systems in regulated, enterprise environments. Proven ability to lead **DevOps modernization, data-intensive backend development, and cross-functional platform initiatives**, ensuring reliability, security, and operational scalability at scale.

PROFESSIONAL EXPERIENCE

Staff Engineer, Addepar

02/2022 – 10/2025
San Francisco, CA

- Architected **scalable portfolio data ingestion pipelines** integrating custodians (Schwab, Fidelity, Pershing) via **REST APIs, SFTP, and event-driven workflows (Kafka)**, supporting multi-billion-dollar AUM environments.
- Designed and implemented high-performance backend services using **Node.js, Python (FastAPI), PostgreSQL, and Redis**, enabling reliable daily **NAV calculation and performance reporting**.
- Led development of an automated **reconciliation engine** with intelligent rule-based matching, reducing manual break resolution by **40%+** and improving operational efficiency.
- Built modular **microservices architecture** for portfolio accounting, fee billing, and rebalancing workflows, improving system scalability and maintainability.
- Delivered API-first services powering **real-time advisor dashboards** and analytics, optimizing query performance and reducing **p95 latency by 35%**.
- Implemented **ML-driven anomaly detection models** to identify transaction inconsistencies and operational risk patterns in financial datasets.
- Strengthened platform security with **SOC 2-aligned controls, structured audit logging, RBAC, and encryption (TLS/KMS)** to protect sensitive client financial data.
- Contributed to **React + TypeScript frontend applications**, delivering interactive portfolio visualizations and advisor reporting tools.
- Modernized CI/CD pipelines using **Docker, Kubernetes, and GitHub Actions**, improving deployment frequency and reducing production rollback time.
- Partnered cross-functionally with Product, Compliance, and Operations teams to translate complex RIA requirements into scalable, production-ready solutions.
- Conducted **architecture reviews and performance tuning initiatives**, ensuring long-term system scalability and reliability.

Solution Architect, SmartBear

07/2015 – 12/2021
Somerville, MA

- Architected and maintained **enterprise-grade CI/CD pipelines** using **Jenkins, GitHub Actions, and Azure DevOps**, automating build, test, security scanning, and deployment workflows across microservices platforms.
- Designed scalable **containerized test execution environments** with **Docker and Kubernetes**, enabling parallel API and performance test runs and reducing regression cycles by 40%.
- Implemented **Infrastructure as Code (Terraform, Helm)** to provision cloud-native test infrastructure on AWS/Azure, ensuring reproducible, version-controlled environments.
- Integrated **automated API contract testing and schema validation** into pull request workflows, preventing breaking changes and improving release stability.

- Built distributed **test orchestration services (Node.js/Python/Java)** to coordinate large-scale test suites across multiple environments and regions.
- Introduced **observability pipelines (Prometheus, Grafana, OpenTelemetry)** to monitor test infrastructure health, pipeline duration, and failure patterns.
- Strengthened DevSecOps posture by embedding **SAST/DAST, dependency scanning, and container image security checks** directly into CI workflows.
- Optimized artifact management and versioning strategies, reducing build times and improving rollback reliability during high-frequency deployments.
- Led migration from legacy VM-based runners to **cloud-native Kubernetes executors**, improving scalability and cost efficiency.
- Partnered with Product and Engineering leadership to drive **release engineering best practices**, improving deployment frequency and lowering change failure rates.

Junior Software Engineer, Rackspace Technology

07/2013 – 06/2015
San Antonio, TX

- Developed and maintained backend services and internal tools using Python and JavaScript to support customer-facing platforms and internal operational workflows.
- Implemented and extended RESTful APIs for service automation, configuration management, and integration with internal systems.
- Assisted in breaking down legacy monolithic services into more modular components to improve maintainability and deployment reliability.
- Supported AWS and private cloud environments, working with compute, storage, and networking services under senior engineer guidance.
- Built automation scripts (Python, Bash) to streamline provisioning, configuration, and routine operational tasks, reducing manual intervention and error rates.
- Assisted with early Infrastructure as Code-style workflows, contributing to scripted deployments and configuration templates before full IaC standardization.
- Worked with relational databases (MySQL, PostgreSQL) to support application features, troubleshoot performance issues, and maintain data integrity.

EDUCATION

Bachelor's Degree of Computer Science, The University of Texas at Austin

09/2009 – 07/2013
Austin, TX

SKILLS

Languages & Runtime

JavaScript (ES6+), TypeScript, Python, Node.js, Golang, Java(JVM), C# (.NET), C++, Rust, SQL, Bash/Shell/ PowerShell, YAML/JSON/Protobuf, GraphQL, CUDA, Julia, WebAssembly (WASM), Ray/Dask, Hugging Face, Inference/OpenAI/VLLMruntimes

DevOps

Terraform, AWS CDK, CloudFormation, Ansible, AWS (ECS, Fargate, Lambda, S3, RDS, CloudWatch, EventBridge), GCP & Azure, CI/CD, Github Actions, Jenkins, GitLab CI, Docker, Kubernetes, Prometheus/ Grafana, ELK, Open Telemetry, CloudWatch, SLO/SLA/ SLIDesign, VPC & Networking, VPN, Zero-Trust, Auto-Scaling

Frontend

JavaScript, TypeScript, React/Next.js, Redux, React Query, SWR, TTI, LCP, CLS, SSR, SSG, SEO, Tailwind CSS, Styled-Component, MUI, Figma, Motion, D3.js, Chart.js, Webpack, Vite, Angular, Vue

AI/ML-LLM Development

Fine-Tuning, RAG, Langchain, Langgraph, Langbase, Lang Smith, Hugging Face, Autogen, CrewAI, DSPY, Open AI Llama 3, PyTorch, Tensorflow, Ray, MLflow, Pinecone, pgvector, FAISS, Multi agent system, Prompt Engineering & Evaluation, SageMaker, GCP Vertex AI, CUDA

Backend & API

Microservices & Distributed Systems, Rest, GraphQL, gRPC, Node.js (Express, NestJS), Python (Flask, FastAPI, Django), Golang (Gin, Fiber), .NET Core/Java Spring Boot, OAuth2, JWT, SSO, RBAC/ABAC, Prisma, TypeORM, SQLAlchemy, EF, Gorm, Redis, Kafka, RabbitMQ, AWS SQS, Websocket, WebRTC, SSE, Pub/Sub, AWS API gateway, NGINX, Kong, Cloudflare

Testing

TDD, BDD, Pytest/unittest (Python), Jest/Mocha/Vitest, xUnit/Nunit, Cypress, Selenium, Playwright, Model Evaluation Pipelines, Prompt Evaluation, Pandera, k6, Locust, JMeter, OWSAP ZAP